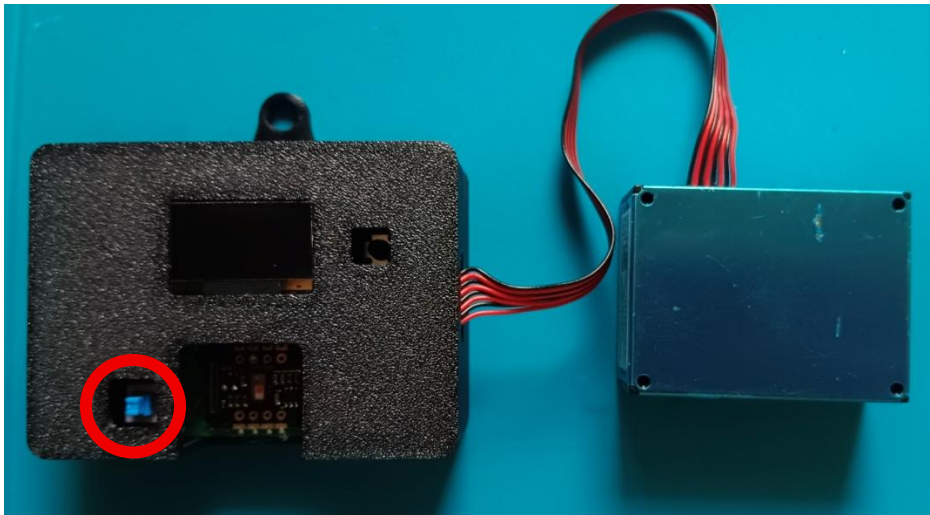


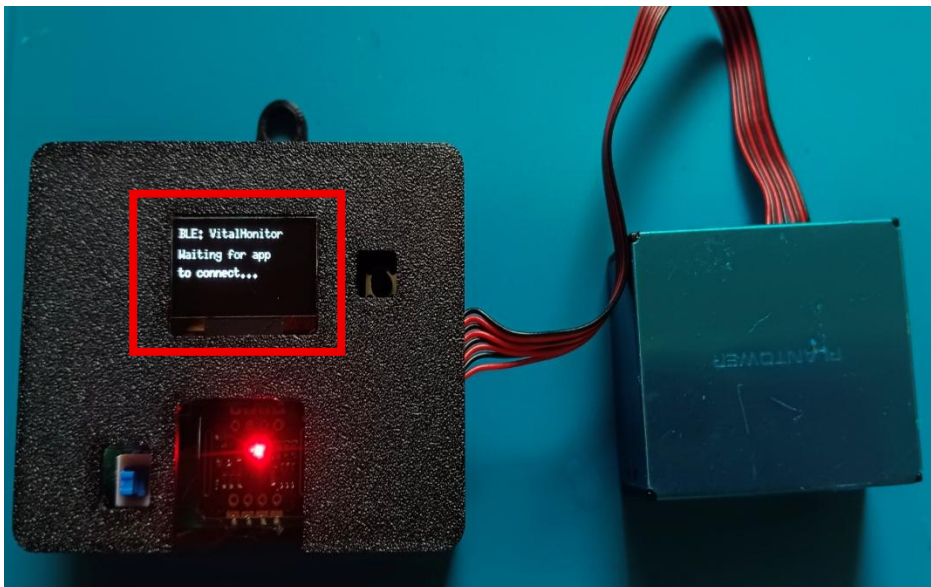
VitalMonitor: full usage steps

A. Power on the device

- A. Power ON the Device by clicking the power button. Give it a few seconds to start up. It first needs to find the MAX30105 sensor. If the sensor is not wired properly, the board halts here and the screen stays blank.

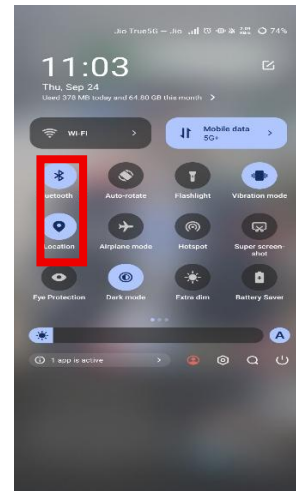
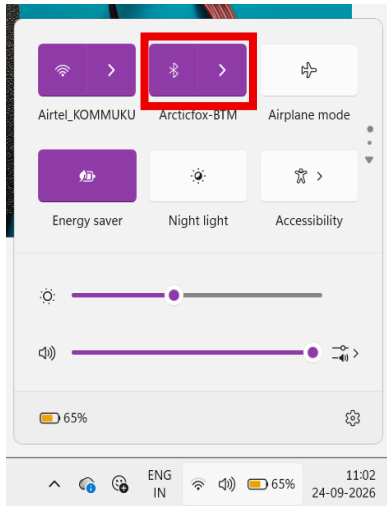


- B. Check the OLED. When it is ready, it shows "**BLE: VitalMonitor / Waiting for app / to connect...**".

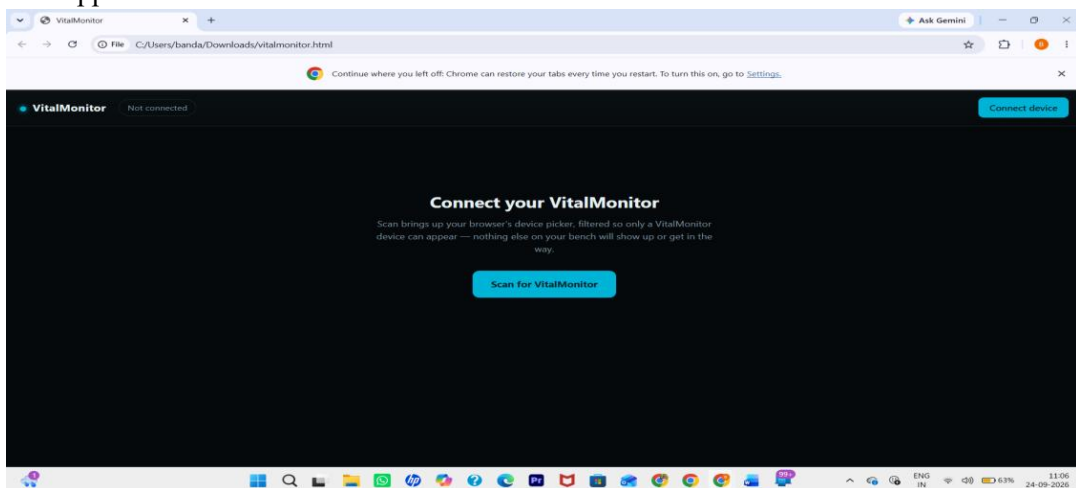


B. Prepare the phone or laptop

1. Turn **Bluetooth ON**.



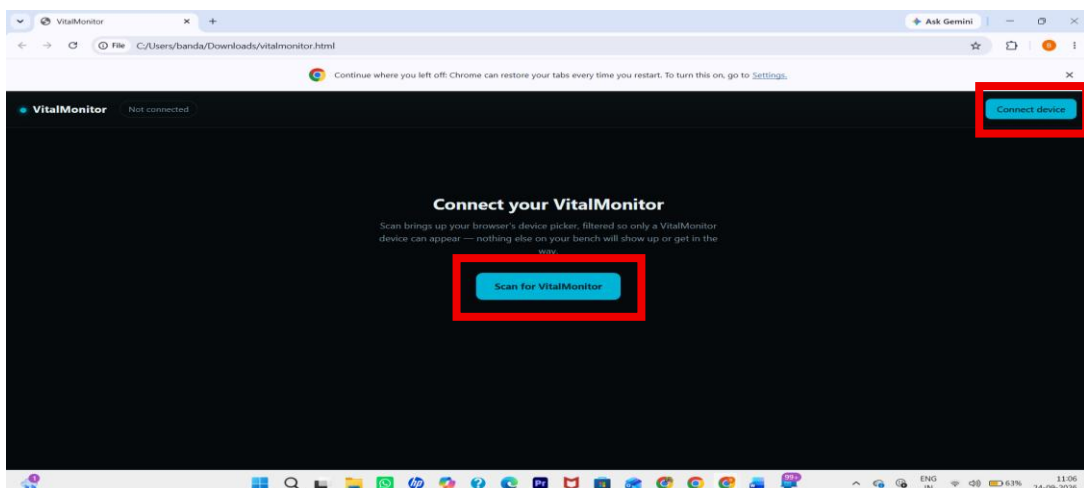
2. On Android, also turn **Location ON**. Chrome will not scan without it.
3. Open the page in **Chrome or Edge**. Safari, Firefox and in-app browsers such as WhatsApp do not support Web Bluetooth.



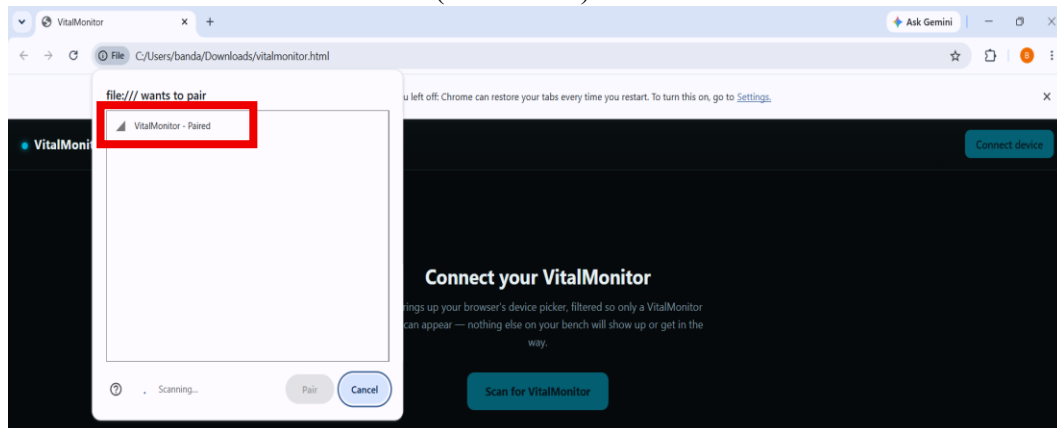
4. Make sure you are **online** the first time. The PDF library loads from the internet.

C. Open the HTML and connect

1. Open vitalmonitor.html in Chrome. You will see "**Connect your VitalMonitor**", and the top bar says **Not connected**.

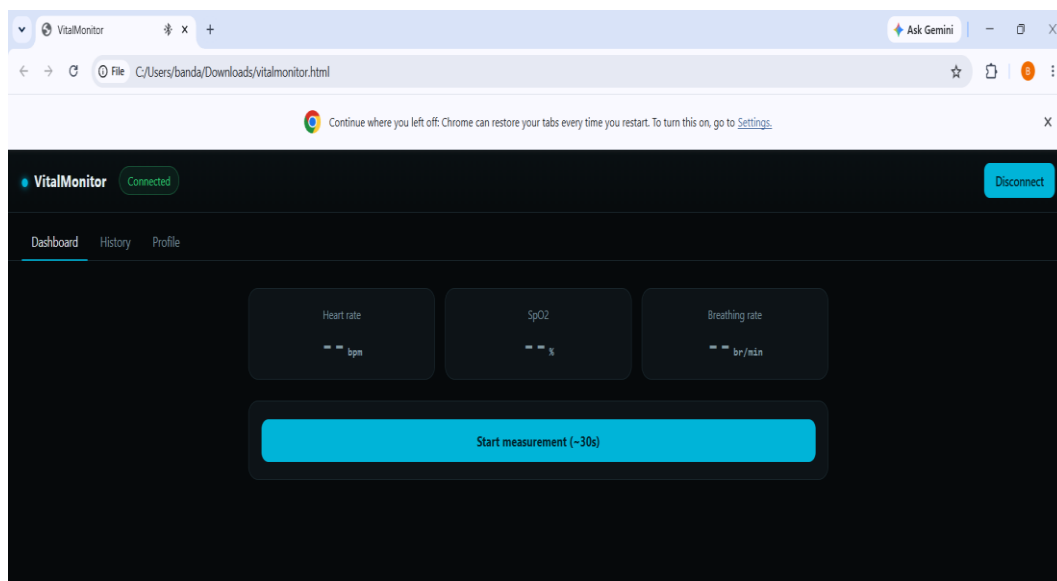


2. Click **Scan for VitalMonitor**. You can also click **Connect device** at the top right.
3. The status changes to **Scanning...**, and a Bluetooth picker pops up.
4. Select **VitalMonitor** and click **Pair** (or **Connect**).



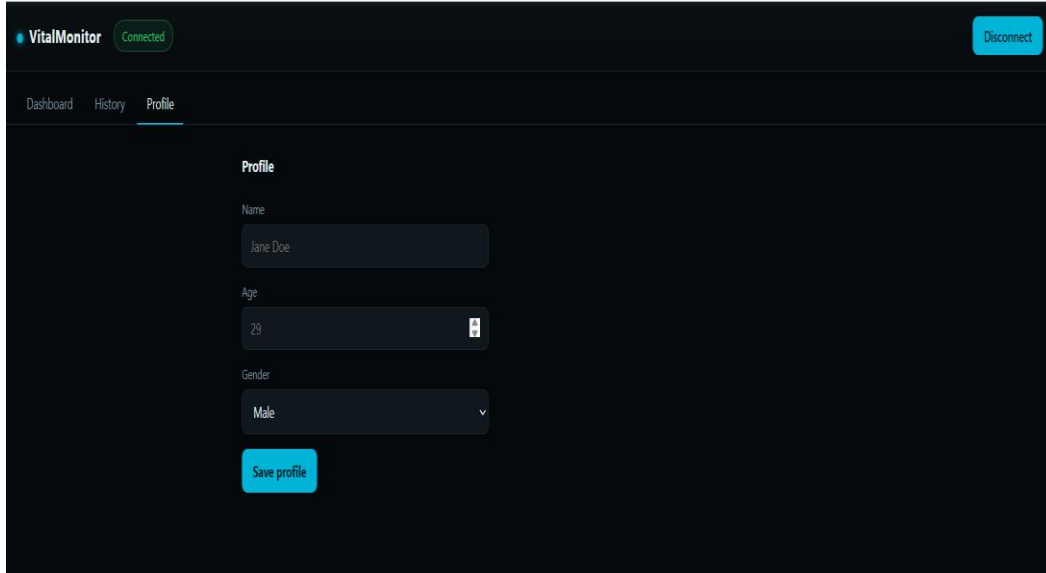
5. The status changes to **Connecting...** and then **Connected**.
6. Check three things to confirm the connection worked:
 - The top bar says **Connected**.
 - The top-right button now says **Disconnect**.
 - The OLED shows "**BLE: Connected**".

The Dashboard, History and Profile tabs also appear. The device clock is set automatically at this point.



D. Fill in the profile (before your first report)

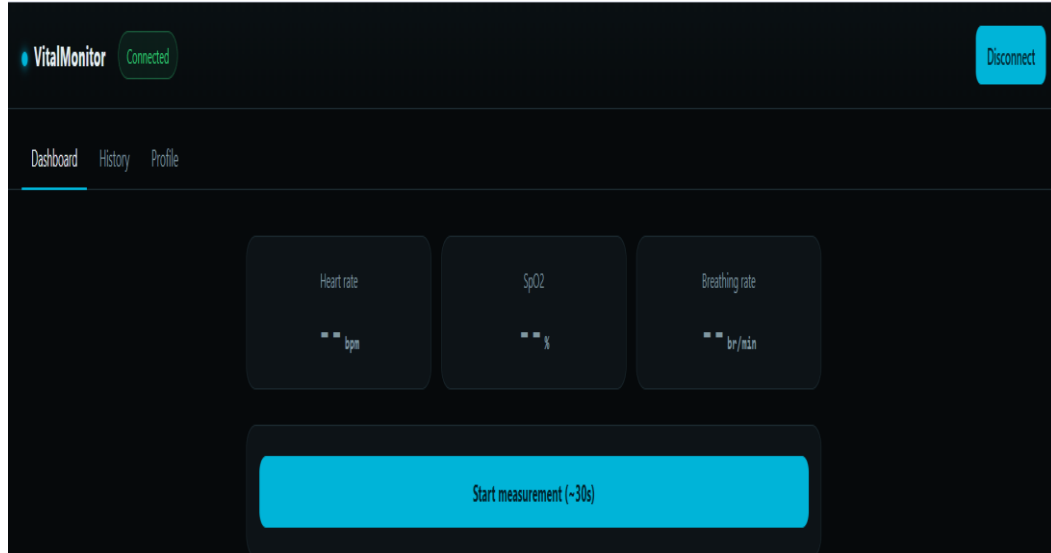
1. Click the **Profile** tab.
2. Enter **Name**, **Age** and **Gender**.
3. Click **Save profile**. A toast message says "Profile saved." This information appears on the PDF reports.



The screenshot shows the VitalMonitor web interface. At the top, there is a header with the VitalMonitor logo, a 'Connected' status indicator, and a 'Disconnect' button. Below the header is a navigation bar with three tabs: 'Dashboard', 'History', and 'Profile'. The 'Profile' tab is currently selected. The main content area is titled 'Profile' and contains three input fields: 'Name' with the value 'Jane Doe', 'Age' with the value '29', and 'Gender' with a dropdown menu showing 'Male'. Below these fields is a blue 'Save profile' button.

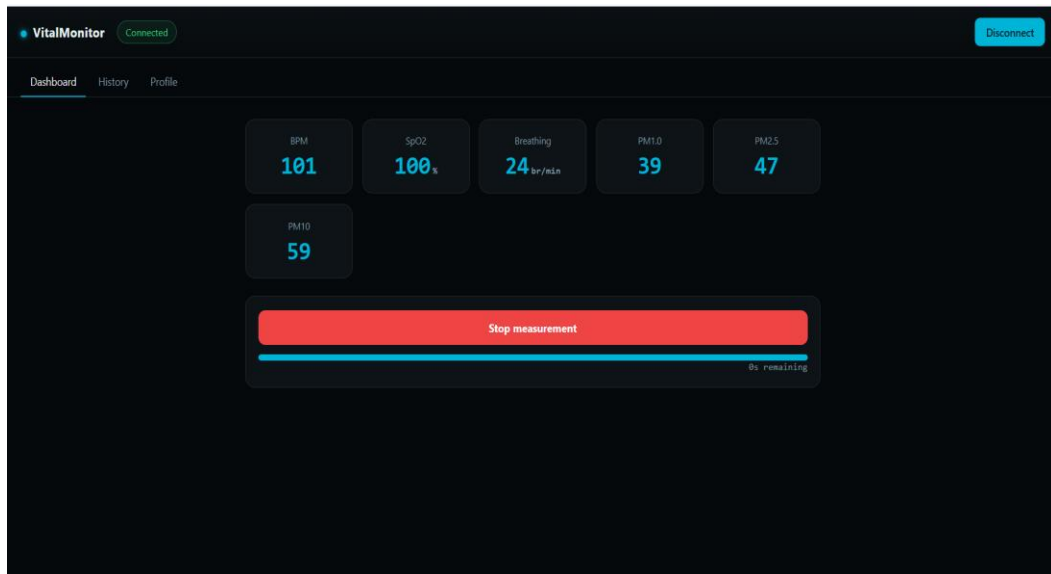
E. Take a measurement

1. Click the **Dashboard** tab.

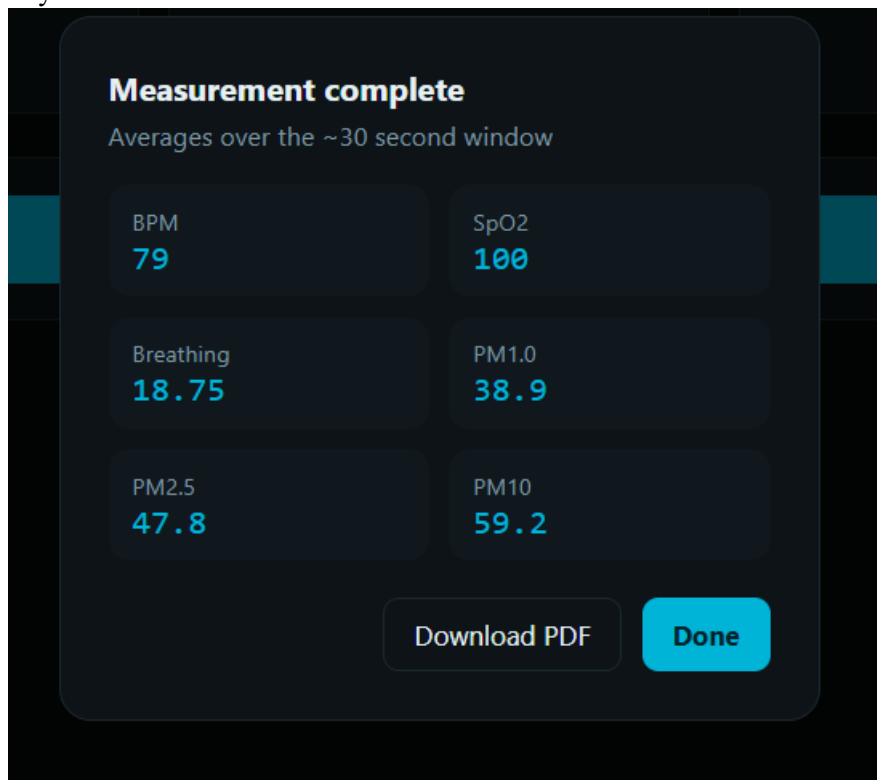


The screenshot shows the VitalMonitor web interface with the 'Dashboard' tab selected. The main content area displays three live measurement cards: 'Heart rate' showing '== bpm', 'SpO2' showing '== %', and 'Breathing rate' showing '== br/min'. Below these cards is a large blue button labeled 'Start measurement (~30s)'.

2. Place your finger on the MAX30105 sensor first. Press gently and hold still. The OLED goes from "Place finger..." to "Stabilizing..." and then shows BPM. This takes about 3 seconds.
3. Click **Start measurement (~30s)**.
4. The page shows six live cards: BPM, SpO2, Breathing, PM1.0, PM2.5 and PM10. A progress bar and a "**30s remaining**" countdown also appear. The OLED shows "Measuring... / Xs left".
5. Keep your finger still for the full 30 seconds. To abort, click **Stop measurement**. An aborted measurement is not saved.

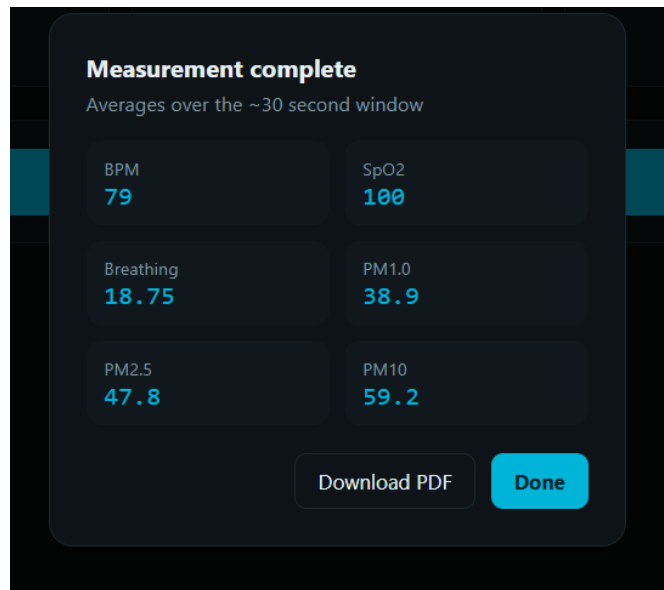


6. After 30 seconds, a **summary popup** shows the averages and the record is saved on the device automatically.

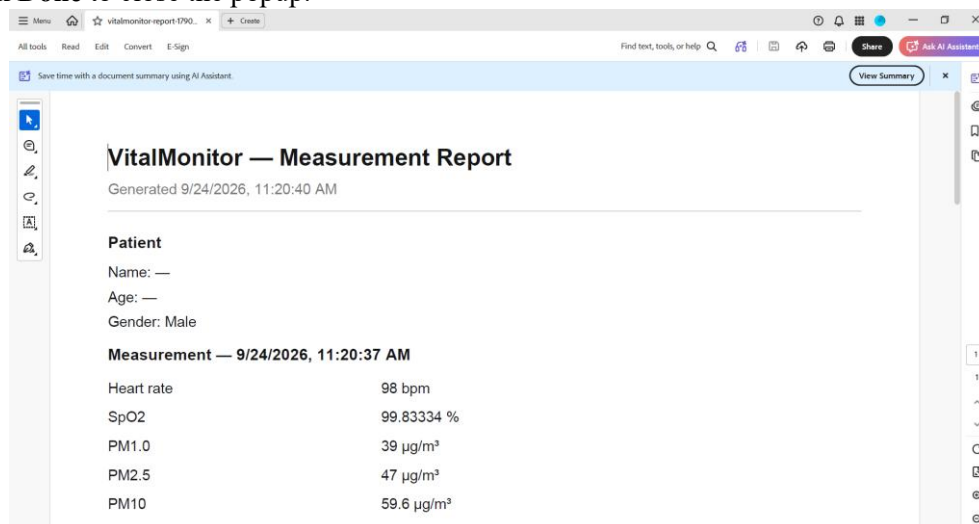


F. Download the single-measurement report

1. In the popup, click **Download PDF**. The report is saved to your Downloads folder as vitalmonitor-report-....pdf.

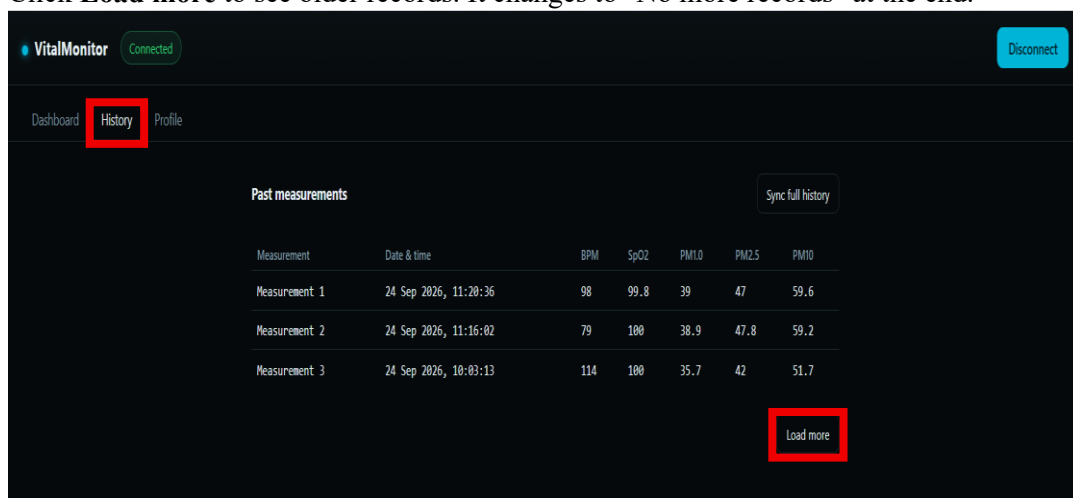


- Click **Done** to close the popup.



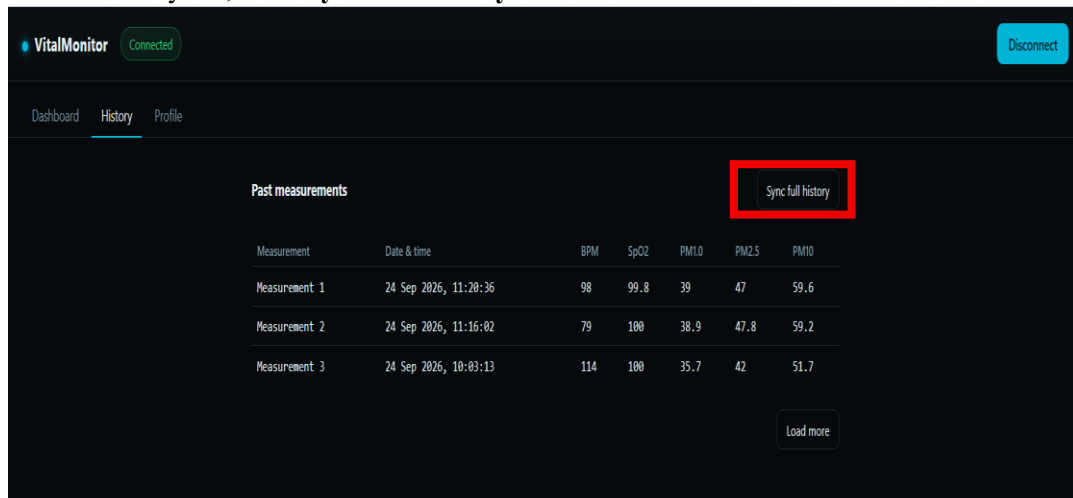
G. View history

- Click the **History** tab.
- The table lists Measurement, Date and time, BPM, SpO2, PM1.0, PM2.5 and PM10.
- Click **Load more** to see older records. It changes to "No more records" at the end.

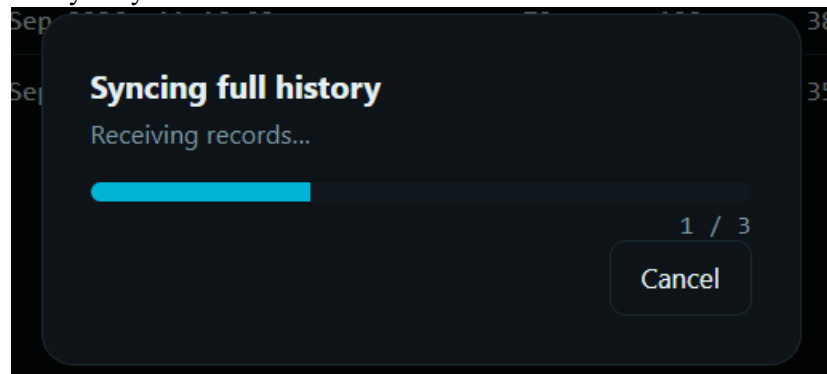


H. Download the full history PDF

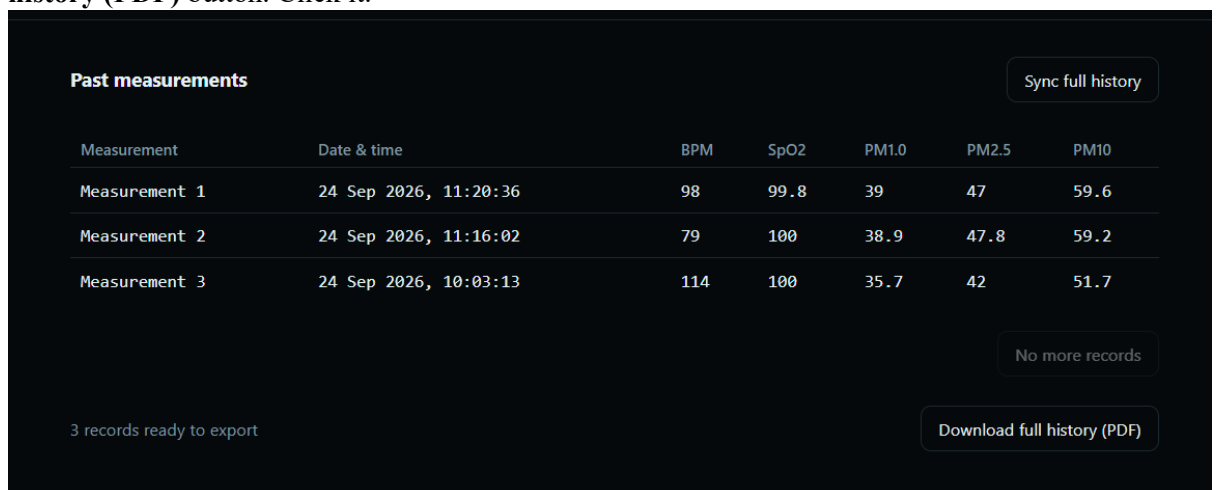
1. On the History tab, click **Sync full history**.



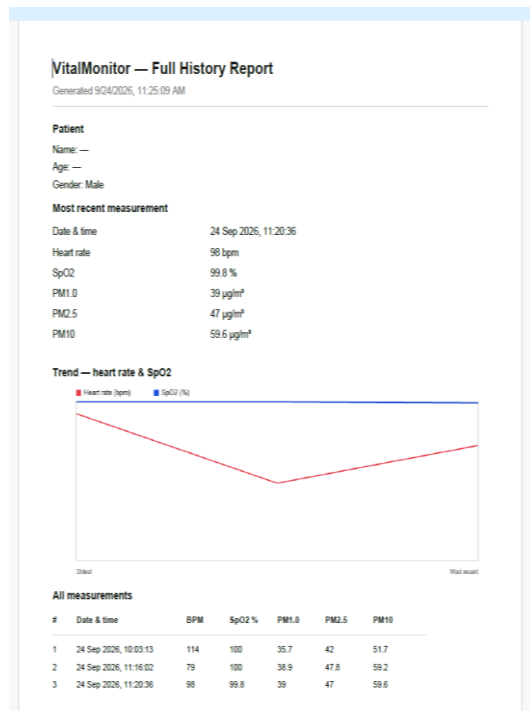
2. A popup shows the progress ("Receiving records..." and a count such as 5 / 20). Wait until it closes. A toast says "Synced N records".



3. A new row appears at the bottom with "N records ready to export" and a **Download full history (PDF)** button. Click it.



4. The file is saved as vitalmonitor-history-....pdf. It has the patient info, the most recent measurement, a trend chart and a table of all measurements.



I. Disconnect

1. Click **Disconnect** at the top right. The status goes back to **Not connected**, and the OLED returns to "Waiting for app".

If something goes wrong

- **"Disconnected — reconnecting..." banner:** move closer to the device. It retries every 2 seconds.
- **The VitalMonitor device is not in the picker:** check that the OLED says "Waiting for app to connect", then press the reset button on the board.
- **Sync fails or times out:** click **Sync full history** again.
- **BPM or SpO2 shows "--" or averages are 0:** the finger was off the sensor or moving. Place it first, then press Start.